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Rotary Hammer Bit Removal Apparatus and Method of Removing a Rotary Hammer Bit

Abstract

An adapter which allows an impact wrench to be coupled to a tool bit for a rotary hammer and thereby rotate the tool bit in either direction. The adapter has a shaft member having a proximate end and a distal end. The proximate end has a rear facing axial opening configured to receive an anvil from the impact wrench. The distal end of the shaft member is coupled with a chuck assembly which is configured to releasably engage a shank of the tool bit and thereby rotate the tool bit. The adapter has a sleeve member which encloses the shaft member and allows the shaft member to rotate within the sleeve member. A handle member may be attached to the sleeve member, wherein the handle member may be utilized to apply a pulling force to the adapter to assist in withdrawal of the tool bit from a foundation.

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Background/Summary

BACKGROUND OF THE INVENTION

[0001] The present invention relates generally to a tool bit removal system, and more specifically to a removal system which recovers tool bits which were stuck while being used with a rotary hammer, also known as a percussion drill. It is to be appreciated that a rotary hammer is to be distinguished from a hammer drill which is less powerful, and which utilizes a different chuck mechanism for retaining a tool bit than the rotary hammer.

[0002] Rotary hammers are heavy-duty tools designed for drilling through concrete, masonry, and the like in construction projects by combining the rotary action of a drill bit with a hammer action. Utilizing a piston driving system, a rotary hammer creates a powerful hammering action which allows the tool bit to break through tough materials. Rotary hammers can be used for chiseling and demolition work, making the devices very useful in construction projects. Rotary hammers are often utilized in specialty applications such as drilling a clearance hole within wood and concrete.

[0003] As compared to a conventional drill and hammer drills, which grip the shank of a drill bit with a three-jaw adjustable chuck to prevent rotation of the bit within the chuck during drilling operations, the “chuck” of a rotary hammer allows the tool bit to float axially within the chuck to simultaneously to allow both the hammering and rotating action, either separately or simultaneously. This chuck is known as the “slotted drive system” or SDS. The SDS chuck is a spring-loaded mechanism of which there are two types SDS-Plus and SDS-Max, with the former suitable for hole size diameters generally ranging from ⅝-inch to ¾-inch and the latter suitable for hole size diameters generally ranging from ⅜-inch to 2-inch. Unlike the drill bits which are used with a conventional drill, which have a smooth shank, the tool bits for rotary hammers have a splined shank which is compatible with the SDS chuck of the rotary hammer, with the tool bit having the nomenclature SDS, SDS-Plus and SDS-Max. These spline shanks lock into the SDS chuck by a spring-biased key or ball bearing which engages a groove of the splined shank, allowing the bit to reciprocate independently of the chuck.

[0004] While very effective in drilling and hammering into a matrix material, a rotary hammer applies the hammering action only while penetrating the matrix material. Some rotary hammers only allow rotation in the forward direction without a reverse direction option. As a result, the rotary hammer may urge the tool bit into a matrix material but not have the functionality to withdraw the tool bit from the material, resulting in the tool bit becoming stuck in the matrix material.

[0005] In one common application, a rotary hammer is used to drill a hole for setting an anchor for attaching a fixture to a cured concrete foundation adjacent to a rebar-reinforced retaining wall. To comply with the applicable building codes, the anchor hole typically must be drilled to a depth ranging from 2″ to 60″. In this situation, there can be several layers of rebar within the retaining wall and extending into the foundation, which creates a high likelihood that the drill bit intersects the rebar during the downward drilling process. When this occurs, metal shavings from the rebar can cause the drill bit to bind up and get stuck. Because the rotary hammer provides no hammer action to facilitate coming out of a hole, and because many hammer drills do not have a reverse direction option, removal of the drill bit from the foundation can be a substantial problem. The problem can be exacerbated because there may be limited working space around the stuck tool bit because of the pre-installed structural members surrounding the location. A stuck tool bit can cause costly project delays and even result in design modifications which require the resubmission of plans for further review and approval.

SUMMARY OF THE INVENTION

[0006] Embodiments of the present invention provide an apparatus which provides a solution to the above-identified problem. Embodiments of the disclosed apparatus provide an adapter which allows a user to connect a pneumatically, hydraulically or electrically powered impact wrench to the splined shaft of the rotary hammer tool bit thereby providing the user to apply both forward (clockwise) and reverse (counterclockwise) directions to the rotary tool bit. The impact wrench provides the user the ability to apply repeated forward and reverse rotational impact with substantial torque to the rotary hammer tool bit to break it free.

[0007] Relative to a vertical orientation, the adapter comprises an upper end and a lower end. The upper end comprises a sleeve member and an internal shaft member disposed within the sleeve member. The internal shaft member has a receptacle configured to receive the anvil of the impact wrench. The lower end of the adapter comprises has a chuck mechanism coupled to the internal shaft member, where the chuck mechanism is configured to receive and releasably attach to the spline end of the tool bit.

[0008] The chuck mechanism may be configured to be actuated by applying a reciprocating motion to a grip assembly of the lower end, where the motion causes an internal key mechanism to either grasp or release the spline end of the tool bit by the interaction of key members of the key mechanism with a wedge or step structure axially disposed within the interior of the grip assembly. As the grip assembly is reciprocated, each key member has an outward facing surface which interacts with an inward facing surface of a wedge member of the wedge structure, where the interaction either causes the key member to be urged inwardly toward the tool bit or allows the key member to move outwardly away from the tool bit. When the position of the key members with respect to the inward facing wedge members allows outward movement of the key members away from the tool bit, biasing mechanisms, such as internal springs, may assist in the disengagement of the key members from the tool bit.

[0009] The grip assembly may be configured so that the chuck mechanism locks onto the shank of the tool bit by reciprocation in a downward motion or in an upward motion. The grip assembly is axially biased by a heavy-duty spring to retain the chuck mechanism in a locked configuration.

[0010] Alternatively, the chuck mechanism may be actuated by manual compression of the grip assembly which causes the engagement of the key members with the spline structure of the tool bit. Once the key members have engaged the spline structure, fasteners or other mechanisms are tightened to maintain the grip assembly in the compressed configuration with the key members affirmatively locked into the spline structure of the tool bit.

[0011] In all embodiments of the invention, the shaft member may freely rotate within the sleeve member and the grip assembly while the chuck assembly may freely rotate with the grip assembly. All embodiments of the apparatus may further comprise an extending handle assembly which attaches to the sleeve member. The extending handle assembly allows a user to apply upward force on the tool bit as the torque is being applied to the tool bit.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] FIG. **1** depicts a situation in which embodiments of the present invention may provide a solution for recovery of a stuck rotary hammer bit.

[0013] FIG. **2** shows different views about the exterior circumference of a shank of a known tool bit utilized with rotary hammers, showing an SDS-Max spline pattern utilized on the shank of the tool bit, with FIG. **2***a* showing an oblique view of the protruding axial key member and an oblique view of one of the elliptically-shaped notches, FIG. **2***b* showing a front view of the protruding axial key member, FIG. **2***c* showing a front view of the axial keyway opposite the protruding axial and

the relative positions of the elliptically-shaped notches on opposite sides of the shank, and FIG. **2***d* showing a front view of the one of the elliptically-shaped notches of the shank.
[0014] FIG. **3** shows a top view of the shank for the tool bit shown in FIG. **2**.
[0015] FIG. **4** shows an embodiment of the present adapter positioned between a pneumatic wrench on one end and a rotary hammer bit.
[0016] FIG. **5** shows a top view of the adapter of FIG. **4** showing the square drive receptacle which is configured to receive the square anvil of a pneumatic wrench.
[0017] FIG. **6** shows a bottom view of the adapter of FIG. **4** showing the chuck mechanism which is configured to receive and secure to the spline of a tool bit.
[0018] FIG. **7** shows a perspective view of the adapter of FIG. **4**.
[0019] FIG. **8** shows a side view of the adapter of FIG. **4**.
[0020] FIG. **9** shows a sectioned view of the adapter taken along line **9-9** of FIG. **8**, showing the chuck mechanism in a closed position.
[0021] FIG. **10** shows an embodiment of a coupling mechanism which may be utilized for connecting a shaft member of the adapter to a chuck shaft member.
[0022] FIG. **11** shows a sectioned view of the embodiment of the adapter depicted in FIG. **9** showing the chuck mechanism in an open position, ready to receive a tool bit.
[0023] FIG. **12** shows a sectioned view of the embodiment of the adapter depicted in FIG. **9** showing the chuck mechanism in the open position with the shank of a rotary hammer bit positioned in the chuck mechanism.
[0024] FIG. **13** shows a sectioned view of the embodiment of the adapter depicted in FIG. **9** showing the chuck mechanism in a closed position with the shank of a rotary hammer bit locked within the chuck mechanism.
[0025] FIG. **14** shows a second embodiment of the present adapter positioned between a pneumatic wrench on one end and a rotary hammer bit.
[0026] FIG. **15** shows a top view of the adapter shown in FIG. **14** showing the square drive receptacle which is configured to receive the square anvil of a pneumatic wrench.
[0027] FIG. **16** shows a bottom view of the adapter shown in FIG. **14** which partially shows the chuck mechanism which is configured to receive and secure to the shank of a rotary hammer bit.
[0028] FIG. **17** shows a perspective view of the adapter of FIG. **14**.
[0029] FIG. **18** shows a side view of the adapter of FIG. **14**.
[0030] FIG. **19** shows a sectioned view taken along line **19-19** of FIG. **18**.
[0031] FIG. **20** shows the sectioned view of FIG. **19** showing a shank of a rotary hammer bit positioned within the chuck mechanism.
[0032] FIG. **21** shows a second perspective view of the adapter shown in FIG. **14** with the chuck mechanism in an open position.
[0033] FIG. **22** shows a sectioned view taken along line **22-22** of FIG. **21**.
[0034] FIG. **23** shows the sectioned view of FIG. **22** further showing a shank of a rotary hammer bit positioned within the chuck mechanism.
[0035] FIG. **24** shows a sectioned view taken along line **24-24** of FIG. **21**.
[0036] FIG. **25** shows the sectioned view of FIG. **24** further showing a shank of a rotary hammer bit positioned within the chuck mechanism.

DETAILED DESCRIPTION OF THE INVENTION

[0037] FIG. **1** depicts a situation in which embodiments of the present invention may be utilized to recover a tool bit **10** which become stuck during a drilling operation with a rotary hammer. As illustrated in FIG. **1**, tool bit **10** having a shank with spline **12** was being used for creating an anchor hole in a cured concrete foundation F adjacent a rebar-reinforced retaining wall W. Vertical framing members S and footing members B have already been disposed upon the foundation F, so it is to be appreciated that if tool bit **10** becomes stuck during the rotary hammer operation, getting the tool bit **10** free presents a substantial challenge because of the limited working room and the

lack of an effective reverse mode in most rotary hammers.
[0038] FIGS. **2**a-**2**d of FIG. **2** show different views about the shank **12** of an SDS-Max tool bit **10** utilized with a rotary hammer. The shank **12** of the SDS-Max tool bit **10** has an axial keyway **14** on one side, an extending axial key **16** on the side opposite the axial keyway, and elliptically-shaped notches **18** on opposing sides of the shank **12**.
[0039] FIG. **3** shows a top view of the tool bit depicted in FIG. **2**, thereby showing the splined shank **12** of the tool bit. Embodiments of the present adapter are configured to securely attach to spined end **12** of an SDS-Max and other known rotary hammer bits and thereby provide an operable couple between an impact wrench **1000** and the stuck tool bit **10**, thereby allowing the tool bit to be rotated in a reverse and forward direction as required to dislodge the tool bit **10** from a matrix such as concrete foundation F depicted in FIG. **1**. Impact wrench **1000** may be actuated pneumatically, hydraulically or electrically. The term “impact wrench” utilized herein shall be understood to include all three power sources.
[0040] FIG. **4** depicts an embodiment of the presently disclosed adapter **100** positioned between a tool bit **10** and an impact wrench **1000**. The impact wrench **1000** is coupled to tool bit **10** by inserting anvil **1002** of the impact wrench into a square drive receptacle **102** of adapter **100** as shown in FIG. **5**. As shown in FIG. **6**, adapter **100** further has chuck mechanism **104** which is adapted to receive and attach to the splined end **12** of the tool bit **10**.
[0041] FIG. **7** depicts a perspective view of an embodiment of the presently disclosed adapter **100** showing the upper end **112** and lower end **114**. FIG. **7** shows the square drive receptacle **102** which is set within an end of shaft member **116**. Once tool bit **10** has been coupled to impact wrench **1000** by adapter **100**, reverse and forward rotation may be applied to the tool bit by the impact wrench. This rotation is applied through shaft member **116** which extends through sleeve member **118** of upper end **112** and partially into lower end **114** where shaft member **116** couples with the chuck mechanism **104** as described below to impart rotational motion to the chuck mechanism and tool bit **10** which is locked within the chuck mechanism. Shaft member **116** may rotate freely within upper end **112** while lower end **114** rotates with the rotational motion of the impact wrench anvil **1002**, turning the tool bit **10**. Upper end **112** may further comprise an extending handle assembly **120** which attaches to sleeve member **118**. Lower end **114** may comprise a grip member **122**.
[0042] FIG. **9**, taken along line **9-9** of FIG. **8**, shows a sectioned view of an embodiment of the presently disclosed adapter **100**. As shown in FIG. **9**, shaft member **116** couples with chuck mechanism **104** by use of a connection pin **124** which is inserted through aligned apertures in the shaft member and the chuck mechanism. The coupling mechanism used between the shaft member **116** and chuck mechanism **104** is configured to provide a joint which allows the application of significant torque without slippage or failure. An example of such an arrangement is depicted in FIG. **10**, showing how shaft member **116** may have an extending member **126** which may engage connection aperture **128** of chuck shaft member **130** with a snug fit and secured by the insertion of the connection pin **124** through the aligned apertures.
[0043] The adapter **100** depicted in FIG. **9** shows detail of the chuck mechanism **104**. FIG. **9** shows the chuck mechanism **104** in a closed position, i.e., the position in which the chuck mechanism would engage and lock onto the splined end **12** of the tool bit **10**, which is not shown in FIG. **9**. Chuck mechanism **104** is considered to be in the closed position because of the interaction between key members **132** with the internal wedges **134** which are axially disposed within the interior of the grip member **122**. In the closed position, as depicted in FIG. **9**, the most inwardly extending portions of wedges **134** are in facing contact with the most outwardly extending portions of key members **132**. As shown in FIG. **10**, chuck shaft member **130** comprises opposite facing elliptical openings **142** through which key members **132** extend to engage the elliptically-shaped notches **18** on opposing sides of the shank **12**.
[0044] The chuck mechanism **104** is biased into the closed position by spring **136**. FIG. **9** also shows an inwardly facing rail member **138**. Rail member **138** engages the axial keyway **14**, shown

in FIG. **2**, of the shank **12** of the tool bit **10**.

[0045] FIG. **11** depicts adapter **100** with the chuck mechanism **104** in an open position, i.e., the position in which the chuck mechanism is ready to slide over the shank **12** of the tool bit **10**. In the open position, as depicted in FIG. **11**, the most inwardly extending portions of wedges **134** are in facing contact with inward recesses of key members **132**, allowing the key members to be radially shifted outward, thereby increasing the diameter of the throat of the chuck mechanism **104**. In order to place the chuck mechanism in the open position, the operator will slide grip member **122** along shaft member **116** against the resistance of spring **136**. Springs **140** assist the key members **132** in moving radially outward, to facilitate the removal of tool bit **10** from the chuck mechanism.

[0046] FIG. **12** depicts adapter **100** with the chuck mechanism **104** in the open position with shank **12** of a tool bit **10** set within the chuck mechanism. As described above, chuck mechanism **104** has been placed in the open position by reciprocating grip member **122** toward upper end **112** thereby compressing spring **136**. As shown in FIG. **12**, key members **132** are extending radially outward away from the shank **12** of the tool bit **10**.

[0047] FIG. **13** depicts adapter **100** with the chuck mechanism **104** in the closed position locked around the shank **12** of tool bit **10**. FIG. **13** shows how key members **132** have been extended into the elliptically-shaped notches **18** on opposing sides of the shank **12** of the tool bit **10**.

[0048] The chuck mechanism is maintained in the closed position by spring **136**. In this position, the anvil **1002** of an impact wrench **1000** may be set within square drive receptacle **102** and the impact wrench may apply rotational motion to shaft member **116** which translates the motion to the chuck mechanism **104** and to the tool bit **10**. Manual upward force may be applied to extending handle assembly **120** to assist in the dislodging of the stuck tool bit **10**.

[0049] An alternative embodiment of the invention is identified as adapter **200**. FIG. **14** depicts adapter **200** positioned between a tool bit **10** and an impact wrench **1000** in the same manner as the above-described embodiment. Similar to the above-described embodiment, impact wrench **1000** is coupled to tool bit **10** by inserting anvil **1002** of the impact wrench into a square drive receptacle **202** of adapter **200** as shown in FIG. **15**. As shown in FIG. **16**, adapter **200** further has chuck mechanism **204** which is adapted to receive and attach to the shank **12** of the tool bit **10**.

[0050] FIG. **17** depicts a perspective view of an embodiment of the presently disclosed adapter **200** showing the upper end **212** and lower end **214**. FIG. **17** shows the square drive receptacle **202** which is set within an end of shaft member **216**. Once tool bit **10** has been coupled to impact wrench **1000** by adapter **200**, reverse and forward rotation may be applied to the tool bit by the impact wrench. This rotation is applied through shaft member **216** which extends through sleeve member **218** of upper end **212** and partially to lower end **214** where the lower end **250** of shaft member **216** couples to chuck plate **252** with bolt **254** to impart rotational motion to the chuck mechanism **204** and tool bit **10** which is locked within the chuck mechanism. Shaft member **216** may rotate freely within upper end **212** while lower end **214** rotates with the rotational motion of the impact wrench anvil **1002**, turning the tool bit **10**. Upper end **212** may further comprise an extending handle assembly **220** which attaches to sleeve member **218**.

[0051] FIG. **19**, taken along line **19-19** of FIG. **18**, shows a sectioned view of the second embodiment of adapter **200**. As shown in FIGS. **18** and **19**, shaft member **216** has a lower end **250** which couples in substantial facing relationship with chuck plate **252** of chuck mechanism **204** and connected together by a high strength bolt **254** such that a strong connection is formed between shaft member **216** and chuck mechanism **204** which is configured to allow the application of significant torque without failure in the connection. High strength bolt **254** has a head which may be disposed within a counter-sunk aperture in lower end **250** and has a substantial length which makes up into threads set within chuck plate **252**. FIG. **19** also shows inwardly facing rail members **238**. Rail members **238** are configured to receive the extending axial key **16**, shown in FIG. **2**, of the shank **12** of the tool bit **10**.

[0052] FIG. **20** depicts a shank **12** of a rotary hammer bit **10** positioned within a closed chuck mechanism **204**. Chuck mechanism **204** comprises sliding key members **256** which translate radially inward and outward on guide pins **258** which extend through slots **264** in each of the slide key members. In the closed position, sliding key members **256** are translated radially inward to each engage the elliptically-shaped notches **18** on opposing sides of the shank **12** of the tool bit **10**. Guide pins **256** extend through cover plate **266**.

[0053] FIG. **21** and the sectional view of FIG. **22** depict the sliding key members **256** translated radially outward on guide pins **258** thereby placing chuck mechanism **204** in an open position and ready to receive the shank **12** of the tool bit **10** as depicted in FIG. **23**. FIGS. **21** through **23** further show the compression rings **260** which, in conjunction with fasteners **262**, are utilized to apply a radially inward force to the sliding key members **256** and thereby close the chuck mechanism **204** about the shank **12**.

[0054] The sectioned views of FIGS. **24** and **25** show a different view of the chuck mechanism **204** showing further detail of the joint between lower end **250** of shaft member **216** and chuck plate **252**, showing the substantial engagement of high strength bolt **254** into chuck plate **252**. FIG. **24** depicts the chuck mechanism **204** in a position ready to receive the shank **12** of the tool bit **10** while FIG. **25** depicts the shank **12** of a tool bit disposed within the chuck mechanism. It is to be appreciated that the chuck mechanism **204** is configured so that substantial engagement force may be applied by the chuck mechanism to the shank **12**.

[0055] Having thus described the preferred embodiment of the invention, what is claimed as new and desired to be protected by Letters Patent includes the following:

Claims

1. An adapter for use in combination with an impact wrench, the adapter configured to couple the impact wrench to a splined shank of a tool bit for a rotary hammer, the adapter comprising: a shaft member comprising a proximate end and a distal end, the proximate end comprising a rear facing axial opening configured to receive an anvil from the impact wrench; a sleeve member which encloses a first portion of the shaft member allowing the shaft member to rotate within the sleeve member; and a chuck assembly coupled to the distal end of the shaft member, the chuck assembly configured to rotate as the shaft member rotates, the chuck assembly further comprising a forward end comprising an axial opening configured to receive the splined shank of the tool bit, the chuck assembly further comprising a key member configured to releasably engage the splined shank of the tool bit such that when the key member is engaged with the spline shank, the tool bit rotates as the chuck assembly rotates.

2. The adapter of claim 1 further comprising an extending handle member attached to the sleeve member.

3. The adapter of claim 1 wherein the chuck assembly comprises a chuck shaft member attached to the distal end of the shaft member.

4. The adapter of claim 3 wherein the chuck assembly comprises a grip member which encloses the chuck shaft member.

5. The adapter of claim 4 wherein the chuck assembly comprises a pair of opposite facing key members which are configured to engage the splined shank of the tool bit upon an interaction of each key member with a corresponding inwardly extending wedge disposed within the grip member.

6. The adapter of claim 5 wherein the pair of opposite facing key members are configured to release the splined shank of the tool bit upon an application of a reciprocal motion to the grip member which shifts the grip member from a locked position, in which the key members are engaged with the splined shank, to a released position, in which the key members are disengaged from the splined shank.

7. The adapter of claim 6 further comprising a spring which biases the grip member in the locked

position.
8. The adapter of claim 7 further comprising a plurality of springs which urge the key members away from the splined shank of the tool bit when the grip member is in the released position.
9. The adapter of claim 1 wherein the chuck assembly comprises a pair of opposite facing key members configured to translate radially inward to engage the splined shank, the key members configured to be releasably retained in a locked engagement with the splined shank.
10. The adapter of claim 9 wherein the key members are retained in the locked engagement by a compression ring which encircles the pair of opposite facing key members.
11. The adapter of claim 9 wherein each of the opposite facing key members comprises a radially oriented slot and each of the opposite facing key members attach to the chuck assembly by a guide pin inserted through the radially oriented slot of each of the opposite facing key members.
12. The adapter of claim 11 wherein the shaft member comprises a lower end and the chuck assembly comprises a chuck plate which attaches to the lower end of the shaft member in a substantial facing relation, wherein the opposite facing key members are attached to the chuck plate by the guide pins.
13. The adapter of claim 12 wherein the chuck plate is attached to the lower end of the shaft member by a bolt.
14. A method of removing a rotary hammer tool bit from a foundation, the method comprising the following steps: disengaging the rotary hammer from the rotary hammer tool bit; attaching an adapter to a spline of the rotary hammer tool bit, wherein the adapter comprises a shaft member comprising a proximate end and a distal end, the proximate end comprising a rear facing axial opening configured to receive an anvil from an impact wrench, the adapter further comprising a sleeve member through which the shaft member extends and freely rotates, the adapter further comprising a chuck assembly configured to releasably attach to the spline of the rotary hammer bit; inserting the anvil of the impact wrench into the rear facing axial opening of the shaft member of the adapter; and engaging the impact wrench to apply a counter-clockwise rotation to the rotary hammer bit.
15. The method of claim 14 wherein an extending handle member is attached to the sleeve member and an upward force is applied to the handle member.